

Cost-Acceptance Markers for Paid Slurm Jobs

CLOACK Stack -- ARCH Portal

How to authorize paid Slurm submissions across CLI, batch, and REST



1. Overview

Paid allocations require an explicit consent token before Slurm accepts the submit. The controller-side `job_submit.lua` plugin checks for this token on every paid job and rejects when it is missing. Free paths (scavenger, `ssci-*` accounts, free QOS) are short-circuited and do not require the marker.

2. Markers

Marker	Where it lives	Submit path	Who sets it
(interactive prompt)	<code>cli_filter.lua</code> runtime	<code>sbatch/srun/salloc</code> on login node	<code>cli_filter.lua</code> -- shows estimated cost and weekly cap, then writes <code>job_desc.comment = 'accept_cost'</code> on 'y'
<code>--comment=accept_cost</code>	CLI flag -> <code>job.comment</code>	<code>sbatch/srun/salloc</code> , non-interactive scripts	Caller -- bypasses prompt, accepts cost blindly
<code>#SBATCH</code> <code>--comment=accept_cost</code>	Batch script header	Batch (<code>.sbatch</code>) script	Caller -- header line at top of script
<code>job.comment = 'accept_cost'</code>	JSON payload field	<code>slurmrestd</code> POST <code>/slurm/v0.0.42/job/submit/</code>	REST client -- mandatory for paid jobs
<code>ACCEPT_COST=1</code>	Env var in <code>job.environment</code>	<code>slurmrestd</code> , OnDemand custom apps, <code>--export</code>	Caller -- alternative to comment

3. Decision Logic in `job_submit.lua`

1. Is partition/QOS free? (scavenger, `ssci-*`, `is_free` QOS)
-> YES: allow, skip cost gate entirely
2. Does `job_desc.comment` contain 'accept_cost'?
-> YES: allow
3. Does `job_desc.environment` contain `ACCEPT_COST=1`?
-> YES: allow
4. Otherwise: REJECT with `slurm.ERROR`

4. Free Paths (no marker required)

Condition	Why no marker
<code>--partition=scavenger</code>	Always free (preemptible)
Account starts with <code>ssci-</code>	Schmidt-funded; <code>FREE_ACCOUNT_PREFIXES</code> short-circuit
QOS has <code>is_free=True</code> (scavenger, <code>ssci</code>)	Free tier
Default allocation (account = <code><pi></code> , no <code><abbr></code>)	Always scavenger
Allocation without IO Number	<code>is_billable=False</code>

5. Common Pitfalls

Symptom	Real cause
<code>sbatch: error: Batch job submission failed: Invalid account or account/partition combination specified</code>	Generic Slurm fallback. Check <code>/var/log/slurm/slurmctld.log</code> -- the lua line 'accept_cost gate: missing' is the real reason ~90% of the time.
REST submit accepted in dev, rejected in prod	Dev cluster may have <code>enforce_billing=False</code> (<code>qos_config.lua</code>); prod enforces.
Script worked yesterday, fails today	Allocation flipped to paid (IO Number added). Add <code>--comment=accept_cost</code> .



cli_filter.lua prompt never appears

You are inside an existing allocation (SLURM_JOB_ID set). Filter skips to avoid double-prompt; parent's acceptance carries through.

6. Examples by Submit Path

6.1 Interactive sbatch (most common)

```
sbatch --account=jdoe_proj1 --partition=low --qos=normal \
--wrap='hostname'
```

6.2 Non-interactive script (no prompt, accept upfront)

```
sbatch --comment=accept_cost \
--account=jdoe_proj1 --partition=low --qos=normal job.sh
```

6.3 Batch script header

```
#!/bin/bash
#SBATCH --account=jdoe_proj1
#SBATCH --partition=low
#SBATCH --qos=normal
#SBATCH --comment=accept_cost
hostname
```

6.4 REST API (slurmrestd)

```
curl -k -H "X-SLURM-USER-TOKEN: $SLURM_JWT" \
-H "Content-Type: application/json" \
-X POST https://slurmrestd:6820/slurm/v0.0.42/job/submit/ \
-d '{
  "job": {
    "account": "jdoe_proj1",
    "partition": "low",
    "qos": "normal",
    "comment": "accept_cost",
    "script": "#!/bin/bash\nhostname\n"
  }
}'
```

6.5 Env var alternative

```
sbatch --export=ACCEPT_COST=1,ALL \
--account=jdoe_proj1 --partition=low job.sh
```

7. References

- * slurm/conf/job_submit.lua -- controller-side enforcement (cost gate)
- * slurm/conf/cli_filter.lua -- login-node prompt and auto-injection
- * slurm/conf/qos_config.lua -- per-cluster enforce_billing toggle
- * CLAUDE.md, section 'Slurm Cost Enforcement'
- * Memory: feedback_rest_submit_accept_cost